Oil Spill Contextual and Boundary-Supervised Detection Network Based on Marine SAR Images

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***Abstract*— Oil spills have caused serious harm to the marine environment. Remote sensing technology is one of the important tools for marine environment monitoring. Synthetic aperture radar (SAR) has become an important technology for detecting marine pollution. Identifying dark spots is essential for oil spill detection based on SAR images. Dark spots' detection can be achieved using image segmentation techniques. However, natural phenomena, such as waves and currents, can also cause dark spots, resulting in consistently uneven intensity, high noise, and blurred boundaries in oil spill images. In addition, existing oil spill detection models often perform well for large targets but have poor detection accuracy for small targets. To solve the above problems, the oil spill contextual and boundary-supervised detection network (CBD-Net) is proposed to extract refined oil spill regions by fusing multiscale features. To improve the internal consistency of oil spill regions, the spatial and channel squeeze excitation (scSE) block is introduced. In CBD-Net, boundary details are enhanced with optimized edge supervision. In addition, a manually labeled dataset is proposed, Deep-SAR Oil Spill (SOS) dataset, aiming to solve the problem of insufficient existing oil spill detection dataset. Experimental results demonstrate that CBD-Net outperforms other comparative models and is able to extract robust and accurate oil spill regions from complex SAR images. The highest mIoU of 83.42% and the highest F1 score of 87.87% were achieved on the SOS dataset. The CBD-Net model proposed in this article can play a guiding role in the marine oil spill decision support system.**

***Index Terms*— Attention block, deep learning, edge supervision, new oil spill dataset, oil spill.**

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I. Introduction

THE ocean plays an important role in global climate and environmental change [1]. However, as humans develop the ocean more and more frequently, the oceans have been polluted to varying degrees. The oil spill is one of the most widespread, frequent, and harmful marine pollutions. The causes of oil spills include ship accidents, oil rig explosions, pipeline breaks, and illegal oil discharges from ships [2]. Due to the limited self-purification capacity of the ocean, oil spills cause serious damage to the marine ecosystem [3]. Therefore, it is very important to monitor oil spill incidents timely and accurate. This is conducive to improving the efficiency and effectiveness of maritime law enforcement agencies [4].

With the advantages of a wide monitoring range and high acquisition efficiency, remote sensing monitoring technology has gradually become one of the main oil spill detection methods [5], [6]. Compared with other remote sensing detection methods, synthetic aperture radar (SAR) has the imaging capability of all-weather, all-time work, and characteristics of variable viewpoint and multipolarization [7]–[9]. It is now recognized as an ideal sensor for oil spill monitoring [7], [10], [11]. The oil film suppresses the short gravity-capillary waves present at the sea surface itself, thus changing the roughness of the sea surface. The Bragg scattering is damped by the sea surface on SAR images. Compared with the surrounding clean sea surface, the oil spill area will appear as a dark spot on the SAR image [11], [12]. The SAR-based method detects oil film by identifying the dark spots area in the image [13]–[15]. Since some natural phenomena in the ocean may also appear as dark areas on the SAR image, these phenomena are called look-alike phenomena. This can easily lead to misclassification [15]. The work of this article focuses on the detection of dark spots. The detection of dark spots can provide an important foundation for further distinguishing oil spills and look-alikes areas.

Dark spots' detection is achieved by using image segmentation techniques. Oil spill detection methods can be classified as threshold-, edge information-, and machine-learning-based segmentation methods [16]. Threshold-based segmentation methods divide grayscale intervals by setting a threshold, which counts the pixels in the same grayscale interval as the

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same category [2], [15], [17]. Edge information-based methods perform image segmentation by concatenating discrete grayscale pixels to obtain region contours [18], [19]. These algorithms have high requirements for the grayscale distribution of the image. However, SAR images have coherent spot noise, and the gray values in the same target area in the image are highly variable [19]. Machine-learning-based segmentation methods for oil film identification can effectively address the limitations of traditional segmentation methods. It is divided into traditional machine learning techniques and deep learning models. Various traditional machine learning techniques have been applied to detect oil leaks from satellite images, including support vector machines [20], [21], decision trees [7], [22], random forests [23], and artificial neural networks [20], [24]–[26]. However, these methods require the manual design of rules and specification of key hyperparameters and require long processing times [27].

In recent years, deep learning models have achieved great success in extracting high-level feature representations of images [28]–[30]. Fully convolutional networks (FCNs) with the encoder–decoder architecture are widely used for oil spill detection [27], [31], [32]. For example, Krestenitis *et al.* [32] used U-Net, LinkNet, Deeplabv3+, and other semantic segmentation models and found that the DeepLabv3+ model achieved the best overall performance in oil spill detection. To further improve the classification accuracy of oil spill area, Zhang *et al.* [33] set simple linear iterative clustering (SLIC) superpixel segmentation as an input to the CNN model. Jiao *et al.* [34] combined the faster region-based convolutional network (faster R-CNN) method and the Otsu algorithm to process UAV images, enabling automatic and accurate detection and evaluation of oil spills. The instance segmentation models also perform well in oil spill detection. Yekeen *et al.* [35] developed a new deep learning oil spill detection model based on the mask-region-based convolutional neural network (Mask R-CNN) model, which is able to segment objects with high accuracy even when overlapping other objects. Nieto-Hidalgo *et al.* [14] proposed a two-stage CNN for SLAR imagery, in which the first stage performs coarse detection and the second stage obtains the exact pixels of the class. In addition, to accurately detect oil spill areas in SAR images, Yu *et al.* [36] attempted to use adversarial learning for minimizing f-divergence.

Although deep learning methods have performed well in many oil spill detection tasks, they still have the following limitations. First, there are few publicly available SAR oil spill datasets, and deep learning requires large-scale data training. Second, oil films on the ocean surface exhibit very irregular shapes with highly complex boundaries. As shown in Fig. 1, the geometric characteristics of dark spots on multipolarized SAR images are expressed in terms of length, width, perimeter, and shape complexity. Existing neural network-based methods do not target oil spill shapes with extremely irregular shapes to design networks. The segmentation effect is not good for oil spill targets with fuzzy boundaries. In addition, relying only on convolution operators to extract features in local regions cannot capture the global context in the image, while

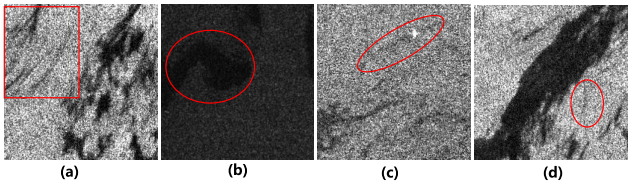


Fig. 1. Four different examples of SAR oil spill images. (a) Oil spill on a bright background. (b) Oil spill on a dark background. (c) Oil spill without a clear boundary on a bright background and with a bright spot near it. (d) Oil spill with a break in the whole oil film.

the contextual information is critical for oil spill detection [37]–[39]. For example, if there is a line black region with sharp bends and no branches, this black region is more likely to be an oil spill area if there is a bright spot nearby. The bright spot is a ship or an oil platform. Otherwise, the linear black area is more likely to be the leeward area of a natural object.

To solve the above issues, a new oil spill detection framework, oil spill contextual, and boundary-supervised detection network (CBD-Net) are proposed to detect oil spills in SAR images. CBD-Net uses the encoder–decoder structure for capturing multiscale oil spills by gradually recovering the spatial information. The parallel spatial and channel squeeze excitation (scSE) blocks [38] are used to selectively collect contextual information and enhance the pixel-by-pixel representation. Compared to previous spatial and spectral attention [39]–[41], the scSE block is more computationally efficient and, thus, is beneficial for real-time oil spill detection. To solve the fuzzy boundaries problem, the joint loss function is designed. The main contributions of this article are given as follows.

1) To improve the feature representation of complex oil spills in SAR images, CBD-Net is proposed. The multiparallel dilated convolution module is inserted in the middle to fuse multiscale feature fusion. The scSE block is introduced to improve the internal consistency of oil spill targets by modeling the global context.
2) The joint loss function is proposed for the blurring and noise occurring inside the object in SAR images. Following the integral method of computing boundary changes, the edge loss term is designed to weigh the binary loss function. This improves the learning ability of edge features in the CBD-Net framework.
3) A new SAR oil spill detection dataset, Deep-SAR Oil Spill (SOS) dataset,[1] is constructed to fully advance the oil spill detection task. The SOS dataset provides a benchmark resource for the development of state-of-the-art algorithms for SAR image oil spill detection or other related tasks.

The rest of this article is organized as follows. Section II describes the details of the SOS dataset. In Section III, the proposed CBD-Net for SAR oil spill detection is introduced. The experimental results and analysis are reported in Section IV. Finally, discussions and conclusions are presented in Sections V and VI, respectively.

[1]The dataset is publicly available online: http://grzy.cug.edu.cn/zhuqiqi/en/yjgk/32384/list/index.htm.

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TABLE I
LIST OF ALOS SATELLITE IMAGES USED IN THIS STUDY

Image	Capture date	Center coordinates
ALPSRP230200560	2010-05-21 04:35:42	lat: 28.734° lon: -90.707°
ALPSRP231220540	2010-05-28 04:24:38	lat: 27.739° lon: -87.821°
ALPSRP231220560	2010-05-28 04:24:54	lat: 28.728° lon: -88.026°
ALPSRP231220580	2010-05-28 04:25:10	lat: 29.723° lon: -88.229°
ALPSRP232970570	2010-06-09 04:20:38	lat: 29.224° lon: -87.056°
ALPSRP232970580	2010-06-09 04:20:46	lat: 29.721° lon: -87.155°
ALPSRP235450560	2010-06-26 04:22:28	lat: 28.728° lon: -87.488°
ALPSRP237930550	2010-07-13 04:24:18	lat: 28.233/° lon: -87.926°
ALPSRP237930560	2010-07-13 04:24:26	lat: 28.728° lon: -88.024°
ALPSRP238660500	2010-07-18 04:30:00	lat: 25.758° lon: -89.029°
ALPSRP238660550	2010-07-18 04:30:41	lat: 28.237° lon: -89.535°
ALPSRP241870520	2010-08-09 04:38:35	lat: 26.749° lon: -91.371°
ALPSRS233043050	2010-06-09 16:25:30	lat: 28.389° lon: -88.302°
ALPSRS239753100	2010-07-25 16:25:45	lat: 25.719° lon: -88.871°

II. SOS: DEEP-SAR OIL SPILL DATASET

A. Study Area 1: Oil Spill Area in the Gulf of Mexico

On April 20, 2010, an explosion on the Deepwater Horizon drilling rig in the U.S. Gulf of Mexico resulted in a major oil spill. The oil slick covered approximately 160 km in length and 72 km at its widest point. The water temperature in the Gulf of Mexico at the time was about 20 °C.

PALSAR is an SAR sensor carried by the ALOS satellite. It uses L-band frequencies to enable cloud-free, and day and night land observations. PALSAR images of the Gulf of Mexico region were acquired between May 2010 and August 2010. The raw data are PALSAR (level 1.5) data products, which have been preprocessed with multivisual processing and map projection operations, with an observation mode of polarimetry and a pixel spacing of 12.5 m [42]. Consistent with the related studies [1], [35], [43], the oil spill is clearly displayed on the HH polarized images. Therefore, the HH polarization of PALSAR is used as the raw data for oil spill detection. The list of ALOS satellite images used in this study is shown in Table I. Fig. 2 shows the location of the oil spill in the Gulf of Mexico and a typical subset of the selected ALOS satellite images. The dark parts of the images were interpreted as crude oil.

B. Study Area 2: Oil Spill Area in the Persian Gulf

The Persian Gulf accounts for more than 50% of the world's oil reserves. Saudi Arabia, Iran, Kuwait, Iraq, and United Arab Emirates along the Persian Gulf coast are important oil-producing countries. The Al-Khafji oil field is an important oil exploration site jointly operated by Kuwait and Saudi Arabia.

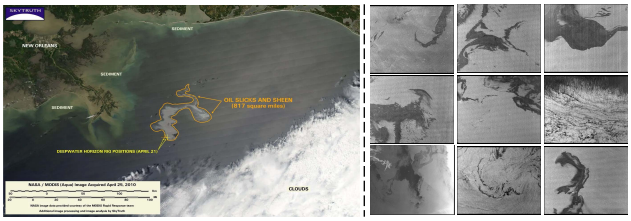


Fig. 2. Location of the oil spill in the Gulf of Mexico. (a) NASA/MODIS satellite image taken April 25 showing oil slicks from Deepwater Horizon disaster (source: John Amos 2010 [44]). (b) Typical subset of the selected ALOS satellite images.

TABLE II
LIST OF SENTINEL 1A SATELLITE IMAGES USED IN THIS STUDY

Image	Capture date	Center coordinates
IW_GRDH_1SDV_01 7782_01DCBB_DE28	2017-08-05 05:12:07	lat: 27.311° lon: 50.597°
IW_GRDH_1SDV_01 7848_01DEC1_EC32	2017-08-09 17:32:09	lat: 28.495° lon: 48.995°
IW_GRDH_1SDV_01 7848_01DEC1_C09A	2017-08-09 17:22:37	lat: 29.002° lon: 48.660°
IW_GRDH_1SDV_01 7855_01DEF7_F48C	2017-08-10 06:41:02	lat: 29.439° lon: 47.535°
IW_GRDH_1SDV_01 7884_01DFD8_8F93	2017-08-12 06:05:57	lat: 27.014° lon: 52.812°
IW_GRDH_1SDV_01 7884_01DFD8_6CAE	2017-08-12 06:05:54	lat: 25.505° lon: 52.485°
IW_GRDH_1SDV_01 7921_01E0ED_DB39	2017-08-14 17:36:49	lat: 28.706° lon: 46.893°

Sentinel 1A is equipped with a C-band SAR sensor with all-weather imaging capability. It can provide measurement data for land, forest, ocean, and glacier monitoring and mapping. In this study, we used Level-1 Interferometric Wide Swath GRD Products. The image width of this mode is 250 km, and the spatial resolution is 5 m × 20 m (monovision). The incidence angle is about 29.1°∼46°. The polarization modes are VV and VH. Consistent with the related studies [1], [35], [43], the VV polarization of Sentinel 1A is used as the raw data for oil spill detection. Then, it performs preprocessing operations, such as speckle filtering, radiation calibration, and topography correction. The Sentinel-1A satellite images used and their parameters are listed in Table II. The location of the oil spill in the Persian Gulf and a typical subset of the selected Sentinel-1A satellite images are given in Fig. 3.

C. Production of SOS Dataset

Existing oil spill detection datasets are difficult to obtain, and there is a lack of public datasets and benchmarks to evaluate the reliability and usability of various oil spill monitoring methods. Therefore, in this article, an oil spill monitoring

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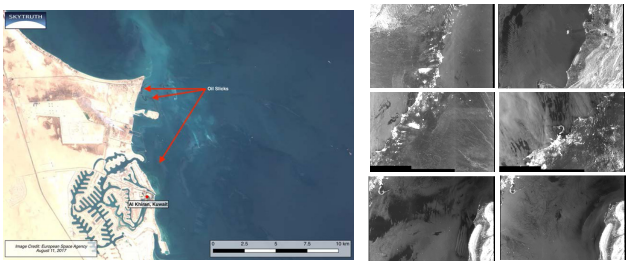


Fig. 3. Location of the oil spill in the Persian Gulf. (a) Sentinel-2A multispectral satellite image showing oil slick making landfall along Kuwait's coast near Al Khiran on August 11, 2017 (source: John Amos 2017 [45]). (b) Typical subset of the selected Sentinel-1A satellite images.

dataset (SOS dataset) is established to evaluate the accuracy of the proposed method and compensate for the lack of oil spill monitoring data sources. As shown in Tables I and II, a total of 21 raw SAR images were collected. The generalization performance of the deep learning model is limited by the diversity of the dataset. For this reason, data enhancement techniques were used to extend the original dataset. Cropping, rotating, adding noise, and other operations are taken on the original grayscale maps. The original 21 SAR images were divided into 6456 oil spill images of 416 × 416 pixels in size. On this basis, sample labels were produced by manual interpretation. The relevant subregions are sampled by GIS experts. In order to eliminate the imbalance between the number of oil spill samples and the number of background samples, the samples with a smaller proportion of oil spill areas were screened. Finally, a total of 8070 labeled oil spill monitoring datasets of 256 × 256 pixels in size were obtained from the two satellite platforms. According to the ratio of 8:2, the datasets were divided into the training set and the test set. A total of 3101 images from the Mexico oil spill area were used for training, and 776 images were used for testing. A total of 3354 images from the Persian Gulf oil spill area were used for training, and 839 images were used for testing.

D. Characteristics of SOS Dataset

First, the SOS dataset has rich image variability. Tolerance for image variation is an important requirement for oil spill monitoring systems. The images in this dataset are carefully selected under a wide range of weather, seasons, lighting conditions, imaging conditions, and scales. Second, the images of the SOS dataset are high in class diversity and interclass similarity. Many performance methods built on deep neural networks have reached saturation in classification accuracy in most existing datasets due to their simplicity or rather lack of variation and diversity. For this purpose, images were obtained under different imaging conditions. Its images were obtained from multiple remote sensing satellite images with a different spatial resolution and spectral coverage.

III. SAR Oil Spill Detection Based on the Proposed CBD-Net Framework

The traditional image segmentation method usually divides an image into several irrelevant regions based on artificially designed features, such as color, spatial texture, and geometry. Inspired by the traditional image segmentation method and characteristics of oil spill targets, the oil spill CBD-Net, which is an end-to-end convolutional neural network, is proposed for oil spill monitoring tasks. The CBD-Net goes through several downsampling layers from input to output, thus gradually obtaining the semantic features of the spill oil at high levels. The semantic features of the higher layers are near the output, whereas the features of the lower layers are near the input. In CBD-Net, the low-level features include features, such as the contours, edges, colors, textures, and shapes of the oil spill, while the high-level features are shown as oil spill-like targets visible to human eyes. To enhance the global contextual representation of oil spill features, the scSE block is embedded to model spectral and spatial attention. In addition, to solve the weak boundary problem of oil spill targets in SAR images, the joint loss function is designed to enhance the network representation of edge features. The network structure of CBD-Net is shown in Fig. 4.

A. Encoder With Contextual and Detailed Information

As shown in Fig. 4, the encoder part is based on the pretrained ResNet-34 [46]. Using the ImageNet pretrained ResNet-34 as initialization can enhance the representation and generalization ability of the CBD-Net, while the convergence speed of the network can be greatly improved during the training process. In Fig. 4, the number in front of the Res-blocks represents the number of Res-blocks, and each Res-block consists of two layers of 3 × 3 Conv. Thus, ResNet-34 has a total of 34 layers of 3 × 3 Conv. Res-blocks have been shown to facilitate the deepening of the network and greatly exploit the potential of deep networks [46]. All convolutional layers are subjected to batch normalization (BN) and linear rectification function operations. The first convolutional layer and the last convolutional layer decrease the number of channels and increase the number of channels, respectively. In the residual operation, the input features and the features obtained from the convolution operation are added directly at each position. The encoder is downsampled by four blocks so that the size of the feature map output from the encoder is reduced to 1/16 of the original image.

To address the high noise and internal inconsistency of the oil spill target, each residual block is followed by an attention module. The scSE block [38] is used to selectively collect contextual information and enhance the pixel-by-pixel representation.

1) scSE Attention Block: As shown in Fig. 5, the feature map $V = [v_1, v_2, v_3, \ldots, v_c] \in R^{C\times H\times W}$ is input to the scSE attention block.

One of the branches performs spatial squeeze and channel excitation (cSE) block. First, the global average pooling is used to compress the spatial dimension on the feature V to generate the feature map $q \in R^{C\times 1\times 1}$. Its kth element is calculated by

$$q_k = \frac{1}{H \times W} \sum_{i}^{H} \sum_{j}^{W} v_k(i, j). \tag{1}$$

This is followed by the excitation operation, which generates weights for each feature channel. The recalibration vector

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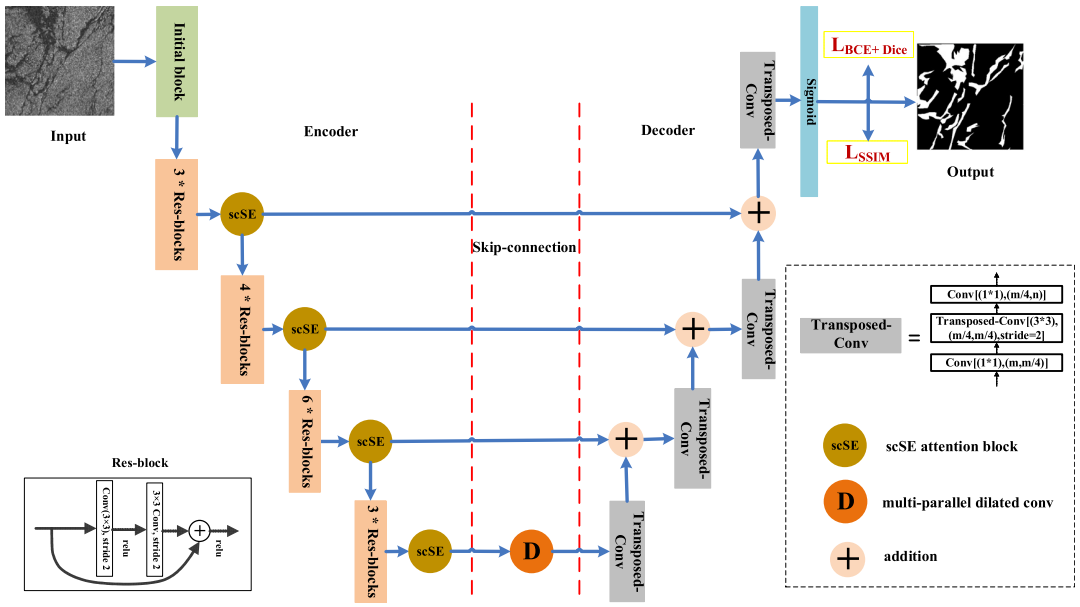


Fig. 4. Overall flowchart of the proposed framework for oil spill detection.

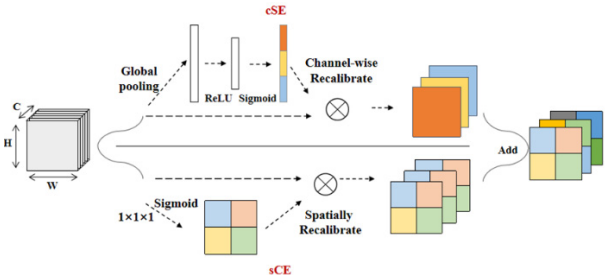


Fig. 5. Overall of the scSE attention block.

is calculated by

$$\hat{q} = W_1(\delta(W_2 q)) \tag{2}$$

where $W_1 \in R^{(C/2)\times C}$ and $W_2 \in R^{C\times(C/2)}$ are the weights of the two fully connected (fc) layers, and $\delta(\cdot)$ denotes the rectified linear unit (ReLU) function. In short, the cSE block is computed as

$$\hat{V}_{\text{cSE}} = [\sigma(\hat{q}_1)v_1, \sigma(\hat{q}_2)v_2, \ldots, \sigma(\hat{q}_c)v_c] \tag{3}$$

where σ denotes the sigmoid. The sigmoid function is used for normalization to obtain the corresponding mask.

Another branch performs channel squeeze and spatial excitation (sSE) block. First, the input feature $V = [v^{1,1}, v^{1,2}, \ldots, v^{i,j}, \ldots, v^{H,W}]$ is squeezed by 1×1 convolution with kernel weight $W_{\text{sq}} \in R^{1\times C\times 1\times 1}$. Here, $v^{i,j} \in R^{C\times 1\times 1}$ corresponding to the spatial location (i, j). We can then write the generated feature as $z \in R^{H\times W}$, where

$$z = W_{\text{sq}} * V. \tag{4}$$

The spatial attention map is then activated using the sigmoid function. The formulation of the sSE block is defined as

$$\hat{V}_{\text{sSE}} = \left[\sigma(z_{1,1})v^{1,1}, \ldots, \sigma(z_{i,j})v^{i,j}, \ldots, \sigma(z_{H,W})v^{H,W}\right]. \tag{5}$$

Finally, the above two blocks are connected in parallel and added together to form the scSE block. $\hat{V}_{\text{scSE}}$ is the final refined output, where

$$\hat{V}_{\text{scSE}} = \hat{V}_{\text{cSE}} + \hat{V}_{\text{sSE}}. \tag{6}$$

B. Multiparallel Dilated Convolution Block

For oil spill detection tasks, small convolution kernels help to detect broken oil film regions. Larger convolution kernels help to eliminate false positive (FP) regions. Multiple dilated convolutions with different receptive fields are merged in parallel. This block effectively ensures the accuracy and detail richness of oil spill detection.

ResNet-34 has five downsampling layers, and if a 256×256 image passes through the encoder section, the output feature map size is 8×8. The dilated convolution kernel $\bar{K}$ is calculated by

$$\bar{K} = R * (K - 1) + 1 \tag{7}$$

where K is the input convolution kernel size and R is known as the dilation rate. The output feature size N is calculated by

$$N = \frac{(I - \bar{K} + 2 * P)}{S} + 1 \tag{8}$$

where I is input feature size, and P and S represent the padding and stride, respectively.

Given that V_n is the edge length of the posterior felt field, V_{n-1} is the edge length of the anterior felt field. For the dilated convolution, the convolutional field is recursively calculated as follows:

$$V_n = V_{n-1} + R^{n-1} * (K - 1). \tag{9}$$

The dilated convolution can be understood as the insertion of $r - 1$ zeros between every two adjacent weights of a convolution with a dilation rate of $R = 1$.

In Fig. 6, dilated convolution with cascaded and parallel modes is included. If a feature map of size $8 \times 8 \times 512$ is given, K is always equal to 3, R_n means that the four values 1, 2, 4, and 8 are taken in different layers, and padding also

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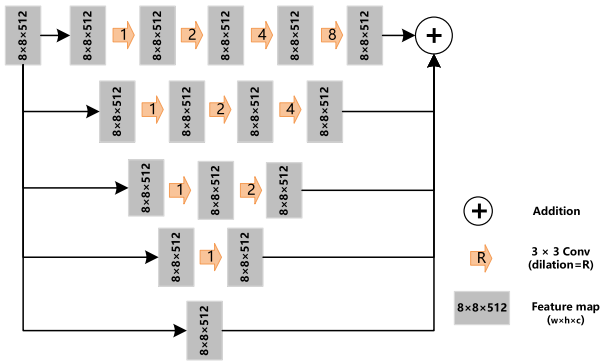


Fig. 6. Overall of the multiparallel dilated convolution block.

takes the values 1, 2, 4, and 8 in different layers. The size of the feature map will remain the same after each layer of the dilated convolution, but the receptive field will be an increasing relationship of 3, 7, 15, and 31. F_n tends to be proportionally amplified overall. These features are, therefore, extracted at different expansion rates, processed on different branches, and then fused to produce the final result. This allows the network to integrate the global semantics while maintaining as much spatial detail information as possible.

C. Decoder

The output of the multiparallel dilation convolution (MDC) module is fed to the decoder. Transposed convolution is used to expand the spatial dimension of the feature map. The decoder has four decoder blocks. The input feature map is first passed through the 1×1 convolution filter, BN, and ReLU of the decoder block. The number of output channels is reduced to 1/4 of the original size. The feature map is then fed into the 1×1 convolution, the transposed convolution layer with a convolution kernel size of 3×3, and a stride size of 2. The feature map is upsampled to twice the original size. The output feature map is then fed into the 1×1 convolution filter, BN, and ReLU.

While the encoder keeps performing downsampling to compress the feature resolution, some useful spatial information will be lost. To recover this information, skip connection is used to fuse the low-level information and the high-level information. As shown in Fig. 4, the output of the scSE block in the encoder is added with the corresponding decoder block through the skip connection. This fusion helps the network to recover more spatial information during upsampling. This is important for fine segmentation.

Through four layers of repeated decoding blocks, the size of the feature mapping gradually increases, and the number of channels gradually decreases. Then, after a 4×4 transposed convolutional layer with a stride of 2 and an ReLU, the feature mapping is restored to the original mapping resolution. Finally, the oil spill area is predicted using the sigmoid function. The output prediction result is the same size as the original image.

D. Joint Loss Function for Edge Preservation

In order to obtain high-quality oil spill segmentation maps with clear boundaries, the joint loss function is proposed. It is defined as

$$L = L_{\text{BCE}} + L_{\text{SSIM}} + L_{\text{Dice}} \tag{10}$$

where L_{BCE}, L_{SSIM}, and L_{Dice} denote the binary cross-entropy (BCE) loss, the structural similarity index (SSIM) loss [47], and the Dice coefficient loss [41], respectively.

L_{BCE} is adopted as the basic loss function. It is defined as

$$L_{\text{BCE}} = -\frac{1}{N}\sum_{i=1}^{N}\left(y_i \log(\bar{y}_i) + (1-y_i)(1-\log(\bar{y}_i))\right) \tag{11}$$

where y_i is the predicted probability of being an oil spill object of pixel i, while $\bar{y}_i$ is the ground-truth label and $\bar{y}_i \in (0, 1)$.

To better interpret boundary pixels, the SSIM loss is used to encourage predictions that respect the structure of the original image. The SSIM loss is sensitive to local structural changes. It assigns a higher weight to the boundary, so the loss is higher near the boundary

$$L_{\text{SSIM}} = 1 - \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{\left(\mu_x^2 + \mu_y^2 + C_1\right)\left(\sigma_x^2 + \sigma_y^2 + C_2\right)} \tag{12}$$

where μ_x, μ_y and σ_x, σ_y are the mean and standard deviations of x and y, respectively, σ_{xy} is their covariance, and $C_1 = 0.01^2$ and $C_2 = 0.03^2$ are used to avoid dividing by zero.

The Dice coefficient loss is proposed for the problem that the proportion of foreground is too small. The dice coefficient is derived from dichotomous classification, which essentially measures the overlap of two samples. L_{Dice} is defined as

$$L_{\text{Dice}} = 1 - \frac{2\sum_{i=1}^{N} y_i \bar{y}_i}{\sum_{i=1}^{N}\left(y_i^2 + \bar{y}_i^2\right)} \tag{13}$$

where y_i is the predicted probability of being an oil spill object of pixel i, while $\bar{y}_i$ is the ground truth.

IV. Experiments and Analysis

A. Implementation Details

The model proposed in this article uses Pytorch as the deep learning framework. All models are trained on four NVIDIA RTX2080 GPUs. The network is optimized, and the parameters are updated using the Adam optimizer [48]. The initial learning rate is set to 2e-4. For better training of the model, the learning rate is automatically adjusted as the number of training epochs increases, which is combined with the optimization to accelerate the convergence of the network. The batch size for the training phase was fixed at 4. Our network undergoes convergence over about 80 epochs. The values after sigmoid output range from 0 to 1. We choose the default 0.5 as the threshold, the prediction probability below the set threshold 0.5 is judged as background, and above the threshold 0.5 is judged as the oil spill target. Of course, choosing 0.4 or 0.6 or other threshold values may give better results.

B. Evaluation Criteria

Several effective indicators, e.g., precision, recall, mean Intersection-over-Union (mIoU), and F1-score, are used to evaluate the performance of different methods. Precision is

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TABLE III
COMPARISON OF CBD-NET WITH OTHER OIL SPILL SEGMENTATION TECHNOLOGIES

	mIoU (%)		F1-score (%)		Recall (%)		Precision (%)	
	PALSAR	Sentinel-1	PALSAR	Sentinel-1	PALSAR	Sentinel-1	PALSAR	Sentinel-1
U-Net	81.63	81.46	82.20	86.10	77.84	81.22	83.08	85.61
D-LinkNet	81.58	82.32	84.57	87.08	85.48	85.22	83.69	85.22
Deeplabv3	81.79	82.94	84.26	87.70	84.31	84.76	83.02	88.08
Ours	**83.31**	**83.42**	**84.84**	**87.87**	**86.75**	**87.32**	**84.20**	**91.20**

the probability of the actual positive samples among all the samples that are predicted to be positive. The recall is the probability that the actual positive samples are predicted to be positive. mIoU is the ratio of the intersection of the set of true values and the set of predicted values. F1-score is a balanced reflection of the accuracy of the algorithm by being able to consider both precision and recall values. True positive (TP) indicates that the model correctly predicts the positive category. FP indicates that the model incorrectly predicts the positive category. False negative (FN) indicates that the model incorrectly predicts the negative category. True negative (TN) indicates that the model incorrectly predicts the negative category. Precision, recall, mIoU, and F1-score are calculated as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{14}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{15}$$

$$\text{mIoU} = \frac{1}{k+1} \sum_{i=0}^{k} \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}} \tag{16}$$

$$\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{17}$$

C. Results and Analysis

The proposed method was compared with U-Net [49], D-LinkNet [50], and Deeplabv3 [51] in two study areas of the SOS dataset, respectively. We compared the qualitative and quantitative analyses of the oil spill areas in the Gulf of Mexico and the Persian Gulf, respectively.

1) Experiment on the Gulf of Mexico Study Area: Fig. 7 shows five representative visualization results on the oil spill images in the Gulf of Mexico. From Fig. 7, we can see that the biggest difficulty of oil spill detection is the high noise and weak boundary of the image. Our results differ from ground truth in some details, such as the area marked by the red box in the last row. Compared with other methods, CBD-Net extracts a more complete oil film and smoother edges. The results of U-Net, Deeplabv3, and D-LinkNet34 are not good for boundary and broken oil film extraction. In Fig. 7, the first row shows a close-up of the broken oil film extraction results, the second row shows the detailed extraction results of the whole oil film, the third indicates the striped oil film, and the fourth to fifth rows show the enlargement of the area where the predicted results do not agree with the ground-truth values. It can be seen that, in the second row, the CBD-Net can extract a more complete oil film because it takes into account the modeling of global contextual relationships.

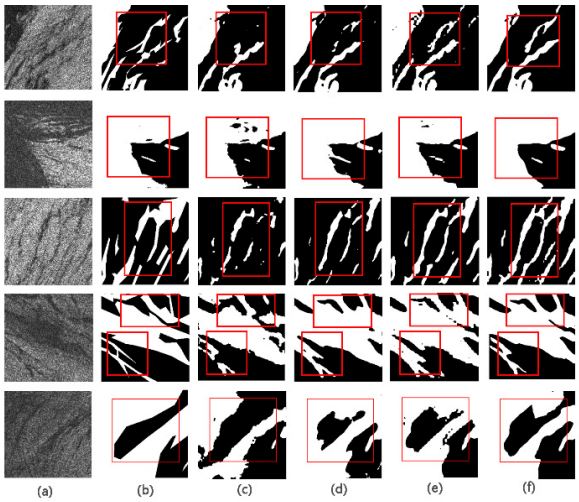


Fig. 7. Oil spill detection results on the Gulf of Mexico study area. (a) SAR image. (b) Ground truth. (c) U-Net. (d) D-LinkNet. (e) Deeplabv3. (f) Ours. The oil spill area is marked in white, and the background is marked in black.

For the third row of fine strips of oil film, the results extracted by other methods are incomplete and fragmented. CBD-Net demonstrates better performance. This indicates that it has solved the multiscale problem of oil spill target to some extent. For the slightly blurred oil film boundary, all four tested methods achieved satisfactory results. As shown in the fifth row, the oil spill boundary on the SAR image is blurred and very similar to the surrounding environment. Without sufficient contextual information, it is difficult to extract it in high quality. However, compared with other methods, the results extracted by CBD-Net are more consistent with the actual oil film boundary. This indicates that CBD-Net can effectively improve the edge accuracy and eliminate the broken oil film.

We used four evaluation metrics, including mIoU, F1-score, precision, and recall to assess the accuracy of the test images in PALSAR acquired images of the Mexico study area. The quantitative evaluation results are shown in Table III. The best results are marked in bold. As can be seen from Table III, the proposed CBD-Net has the highest mIoU and F1-score scores, indicating that our algorithm can maintain a balance between accuracy and recall, and has the highest accuracy. The corresponding precision and recall are 86.75% and 84.20%, respectively, indicating that 86.75% of the detected oil spill pixels are also oil spill pixels in the reference map, and 84.20% of all oil spill pixels in the reference map are correctly detected. This method achieves the best performance in all metrics compared to other methods. In detail, the IoU and F1-scores of CBD-Net are improved by 1.68% and 2.64%, respectively, compared with U-Net, which indicates the effectiveness of

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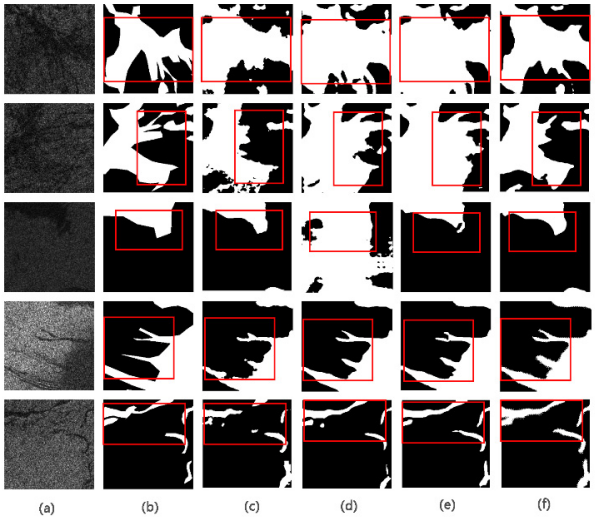


Fig. 8. Oil spill detection results on the Persian Gulf study area. (a) SAR image. (b) Ground truth. (c) U-Net. (d) D-LinkNet. (e) Deeplabv3. (f) Ours. The oil spill area is marked in white, and the background is marked in black.

multiscale contextual aggregation. Compared with D-LinkNet and Deeplabv3, CBD-Net with global contextual information modeling improves the mIoU by 1.73% and 1.52%, respectively. These indicate that the oil spill boundaries extracted by our method are homogeneous, and fewer oil spill regions are detected incorrectly.

2) Experiment on the Persian Gulf Study Area: To further test the performance of CBD-Net under various complex oil spill scenarios, we conducted experiments on the Persian Gulf oil spill images. The visualization results of the comparative experiments are shown in Fig. 8. It can be seen from Fig. 8 that the CBD-Net proposed in this article visually outperforms other comparative methods in dealing with various complex oil spills and extracts the most complete oil spill area. From the first three rows of Fig. 8, it can be seen that, when there is little difference in grayscale between the target and the background, the comparison model is difficult to identify a clear oil spill outline from the image. This can make the judgment of the oil spill target seriously wrong, as shown in column d of the third row. Our method is less influenced by the background because it considers modeling the spatial context and can maintain the regular oil spill shape better. The narrow oil spill shape is shown in the red box in the fourth row. It can be seen that our method can have good extraction for both large- and small-scale oil spill areas. All other tested methods fail to detect high-quality oil spill targets in the blocked area. The fifth row in Fig. 8 shows the extraction results of the broken oil film. It can be seen that the results of U-Net, D-LinkNet, and Deeplabv3 perform poorly in terms of completeness and correctness. Since our method takes into account the boundary features and global context modeling, a complete oil spill region can be obtained.

Table III shows the accurate results of the proposed CBD-Net on the images of the Persian study area obtained by Sentinel-1. It can be clearly seen that the proposed CBD-Net achieves the best performance with 1.96%, 1.77%, 6.1%, and 5.98% improvements in mIoU, F1-score, recall, and precision, respectively. The average accuracy and recall scores of CBD-Net are above 87%, indicating that our method has fewer false detections and missed detections. These illustrate the importance of capturing remote spatial relationships for detecting oil spill objects and also the importance of supervision through enhanced edge features.

TABLE IV
ABLATION STUDY WITH DIFFERENT COMPONENTS COMBINATIONS ON THE PERSIAN STUDY AREA IMAGES

Baseline	MDC	scSE	Joint loss	mIoU(%)	F1-score (%)
√				80.53	82.25
√	√			81.59	83.52
√		√		82.13	84.13
√			√	82.27	84.21
√	√	√	√	**83.31**	**84.84**

TABLE V
ABLATION STUDY WITH DIFFERENT COMPONENTS COMBINATIONS ON THE MEXICAN STUDY AREA IMAGES

Baseline	MDC	scSE	Joint loss	mIoU(%)	F1-score (%)
√				78.79	85.06
√	√			79.16	85.21
√		√		83.35	88.19
√			√	80.09	85.91
√	√	√	√	**83.42**	**87.87**

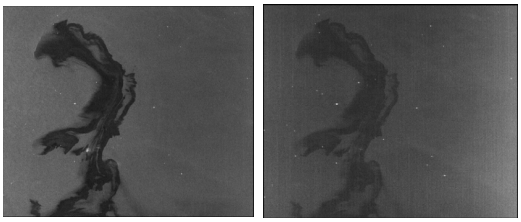
V. DISCUSSION

A. Ablation Study

To better evaluate the contribution of each module, ablation experiments were designed to analyze the impact of each module and different loss functions. The LinkNet-34 [52] is chosen as the baseline. Then, the MDC block and the scSE block are introduced on the baseline, respectively. The BCE + dice coefficient loss in the baseline is replaced by the joint loss function to evaluate the contribution of different loss functions to oil spill extraction. Table IV shows the experimental results on the study area in Mexico. As shown in the second row of Table IV, the performance improves with the introduction of MDC. With MDC, mIoU was improved by 0.66%, and F1-score was improved by 2.3%. The network performance is further improved by adding the scSE block to the network. The mIoU and F1-score are 82.13% and 84.13%, corresponding to increases of 1.6% and 1.88%, respectively. This indicates that the modeling of both multiscale information and global contextual information improves the accuracy of the network. As shown in the fifth row of Table IV, there is a significant improvement in the experiments using the joint loss function, with mIoU and F1-score corresponding to increases of 1.74% and 1.96%. This validates the effectiveness of using the joint loss function in the oil spill monitoring task. As shown in the last row of Table IV, the combination of various modules can improve the comprehensive performance of the network. It is more conducive to extracting oil spill areas in a refined manner.

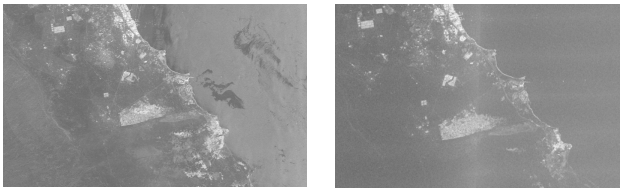
To discuss the effect of different polarization methods and so on on the performance in the next subsection, the same number of samples was selected to control the variables in

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(a) (b)

Fig. 9. (a) HH polarized PALSAR data. (b) HV polarized PALSAR data.



(a) (b)

Fig. 10. (a) VV polarized Sentinel 1A data. (b) VH polarized Sentinel 1A data.

the Persian Gulf study area. As shown in Table V, the performance was improved with the introduction of MDC. After the introduction of scSE, mIoU was improved by 4.56%, and F1-score was improved by 3.13%. These improvements are related to the ability of scSE to improve the integrity of the oil spill target. The experiments using the joint loss function showed significant improvements, verifying the effectiveness of using the joint loss function.

B. Influence of the Polarization

PALSAR images were used in the Gulf of Mexico study area. Sentinel 1A images were used for the Persian Gulf study area. Fig. 9 shows that HH polarization is more advantageous than HV polarization in identifying oil spills. As shown in Fig. 10, oil spills are more easily detected in VV polarization than in VH polarization. The VV and HH polarization images are more suitable for identifying oil spills than cross polarization (HV and VH). Therefore, in this article, PALSAR with HH polarization and Sentinel 1A with VV polarization is chosen to identify oil spills in the ocean. This is consistent with the conclusion reached by Alpers *et al.* [1]. The spatial resolution of PALSAR is 12.5 m, while that of Sentinel 1A is 5 m $\times$ 20 m. As shown in Tables IV and V, the experimental results of Sentinel 1A outperformed PALSAR with the same number of samples, suggesting that Sentinel-1A with the higher resolution has greater potential for observing oil spills in the ocean.

C. Stability of CBD-Net

To evaluate the quality of oil spill extraction, Fig. 11 plots the precision–recall (P–R) curves of the different methods tested on the SOS dataset. The P–R curves give a very visual indication of the precision and recall of the classifier on the sample overall. As shown in Fig. 11, our CBD-Net outperforms the other three comparison methods in terms of P–R curves. Fig. 11(a) shows the P–R curves on the oil spill images in the Gulf of Mexico study area. With increasing recall, the P–R curves of the proposed CBD-Net are higher

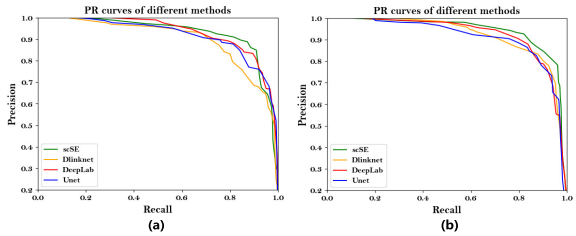


Fig. 11. Illustration of PR curves on the SOS dataset. (a) Gulf of Mexico study area. (b) Persian Gulf study area.

than Deeplabv3 (red curve), D-LinkNet (yellow curve), and U-Net (blue curve) in most cases. Fig. 11(b) shows the P–R curves on the oil spill images in the Persian Gulf study area. It is obvious that the P–R curves of our method are consistently much higher than the PR curves of other comparative methods as the recall rate increases. The results show that the method has good adaptability to SAR images on different sources.

VI. Conclusion

In this article, CBD-Net is designed to extract pixel-level oil spill regions from SAR images. Through the modeling of global context and the fusion of multiscale features, CBD-Net is able to capture multiscale and fine structures for fine-grained oil spill segmentation. Oil spill regions with well-defined boundaries are generated by an improved joint loss function. To evaluate the proposed method, a new oil spill dataset, the SOS dataset, is proposed, and extensive experiments are conducted on it. The experimental results show that the proposed CBD-Net is able to handle the cases, where the difference between the target class and the background class in SAR images is not obvious, and obtain oil spill regions with smooth and complete edges. Future work will be based on a semisupervised approach to alleviate the difficulty of manual labeling of oil spill data. We will also focus more on classifying oil spills and look-alikes in dark spots in complex scenes.

Acknowledgment

The dataset is publicly available online: http://grzy.cug.edu.cn/zhuqiqi/en/yjgk/32384/list/index.htm.

References

[1] W. Alpers, B. Holt, and K. Zeng, "Oil spill detection by imaging radars: Challenges and pitfalls," *Remote Sens. Environ.*, vol. 201, pp. 133–147, Nov. 2017.

[2] S. T. Yekeen and A.-L. Balogun, "Advances in remote sensing technology, machine learning and deep learning for marine oil spill detection, prediction and vulnerability assessment," *Remote Sens.*, vol. 12, no. 20, pp. 1–31, Oct. 2020.

[3] E. Ramsey, A. Rangoonwala, and C. E. Jones, "Marsh canopy structure changes and the deepwater horizon oil spill," *Remote Sens. Environ.*, vol. 186, pp. 350–357, Dec. 2016.

[4] U. Bhangale, S. S. Durbha, R. L. King, N. H. Younan, and R. Vatsavai, "High performance GPU computing based approaches for oil spill detection from multi-temporal remote sensing data," *Remote Sens. Environ.*, vol. 202, pp. 28–44, Dec. 2017.

[5] Y. Liu *et al.*, "Assessment of offshore oil/gas platform status in the northern Gulf of Mexico using multi-source satellite time-series images," *Remote Sens. Environ.*, vol. 208, pp. 63–81, Apr. 2018.

[6] S. Liu and M. E. Hodgson, "Satellite image collection modeling for large area hazard emergency response," *ISPRS J. Photogramm. Remote Sens.*, vol. 118, pp. 13–21, Aug. 2016.

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[7] M. Konik and K. Bradtke, "Object-oriented approach to oil spill detection using ENVISAT ASAR images," *ISPRS J. Photogramm. Remote Sens.*, vol. 118, pp. 37–52, Aug. 2016.

[8] I. K. Lee, A. Shamsoddini, X. Li, J. C. Trinder, and Z. Li, "Extracting hurricane eye morphology from spaceborne SAR images using morphological analysis," *ISPRS J. Photogramm. Remote Sens.*, vol. 117, pp. 115–125, Jul. 2016.

[9] M. M. Espeseth, C. E. Jones, B. Holt, C. Brekke, and S. Skrunes, "Oil-spill-response-oriented information products derived from a rapid-repeat time series of SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3448–3461, Jun. 2020.

[10] D. Latini, F. Del Frate, and C. E. Jones, "Multi-frequency and polarimetric quantitative analysis of the Gulf of Mexico oil spill event comparing different SAR systems," *Remote Sens. Environ.*, vol. 183, pp. 26–42, Sep. 2016.

[11] A.-B. Salberg, O. Rudjord, and A. H. S. Solberg, "Oil spill detection in hybrid-polarimetric SAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 10, pp. 6521–6533, Oct. 2014.

[12] M. Vespe and H. Greidanus, "SAR image quality assessment and indicators for vessel and oil spill detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 50, no. 11, pp. 4726–4734, Nov. 2012.

[13] L. Gemme and S. G. Dellepiane, "An automatic data-driven method for SAR image segmentation in sea surface analysis," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 5, pp. 2633–2646, May 2018.

[14] M. Nieto-Hidalgo, A. J. Gallego, P. Gil, and A. Pertusa, "Two-stage convolutional neural network for ship and spill detection using SLAR images," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5217–5230, Sep. 2018.

[15] Y. Shu, J. Li, H. Yousif, and G. Gomes, "Dark-spot detection from SAR intensity imagery with spatial density thresholding for oil-spill monitoring," *Remote Sens. Environ.*, vol. 114, no. 9, pp. 2026–2035, Sep. 2010.

[16] R. Al-Ruzouq *et al.*, "Sensors, features, and machine learning for oil spill detection and monitoring: A review," *Remote Sens.*, vol. 12, no. 20, p. 3338, Oct. 2020.

[17] P. Liu, C. Zhao, X. Li, M. He, and W. Pichel, "Identification of ocean oil spills in SAR imagery based on fuzzy logic algorithm," *Int. J. Remote Sens.*, vol. 31, nos. 17–18, pp. 4819–4833, Sep. 2010.

[18] D. Cantorna, C. Dafonte, A. Iglesias, and B. Arcay, "Oil spill segmentation in SAR images using convolutional neural networks. A comparative analysis with clustering and logistic regression algorithms," *Appl. Soft Comput.*, vol. 84, Nov. 2019, Art. no. 105716.

[19] W. Wang, H. Sheng, S. Liu, Y. Chen, J. Wan, and J. Mao, "An edge-preserving active contour model with bilateral filter based on hyperspectral image spectral information for oil spill segmentation," in *Proc. Workshop Hyperspectral Image Signal Proces., E*, Sep. 2019, pp. 1–5.

[20] L. Xu, J. Li, and A. Brenning, "A comparative study of different classification techniques for marine oil spill identification using RADARSAT-1 imagery," *Remote Sens. Environ.*, vol. 141, pp. 14–23, Feb. 2014.

[21] D. Mera, V. Bolon-Canedo, J. M. Cotos, and A. Alonso-Betanzos, "On the use of feature selection to improve the detection of sea oil spills in SAR images," *Comput. Geosci.*, vol. 100, pp. 166–178, Mar. 2017.

[22] K. Topouzelis and A. Psyllos, "Oil spill feature selection and classification using decision tree forest on SAR image data," *ISPRS J. Photogramm. Remote Sens.*, vol. 68, pp. 135–143, Mar. 2012.

[23] S. Tong, X. Liu, Q. Chen, Z. Zhang, and G. Xie, "Multi-feature based ocean oil spill detection for polarimetric SAR data using random forest and the self-similarity parameter," *Remote Sens.*, vol. 11, no. 4, p. 451, Feb. 2019.

[24] S.-H. Park, H.-S. Jung, and M.-J. Lee, "Oil spill mapping from Kompsat-2 high-resolution image using directional median filtering and artificial neural network," *Remote Sens.*, vol. 12, no. 2, p. 253, Jan. 2020.

[25] K. Topouzelis, V. Karathanassi, P. Pavlakis, and D. Rokos, "Detection and discrimination between oil spills and look-alike phenomena through neural networks," *ISPRS J. Photogram. Remote Sens.*, vol. 62, no. 4, pp. 264–270, Sep. 2007.

[26] S. Singha, T. J. Bellerby, and O. Trieschmann, "Satellite oil spill detection using artificial neural networks," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 6, no. 6, pp. 2355–2363, Dec. 2013.

[27] F. M. Bianchi, M. M. Espeseth, and N. Borch, "Large-scale detection and categorization of oil spills from SAR images with deep learning," *Remote Sens.*, vol. 12, no. 14, p. 2260, Jul. 2020.

[28] Q. Zhu *et al.*, "A global context-aware and batch-independent network for road extraction from VHR satellite imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 175, pp. 353–365, May 2021.

[29] Q. Zhu *et al.*, "A spectral-spatial-dependent global learning framework for insufficient and imbalanced hyperspectral image classification," *IEEE Trans. Cybern.*, early access, May 25, 2021, doi: 10.1109/TCYB.2021.3070577.

[30] Q. Zhu, Z. Li, Y. Zhang, and Q. Guan, "Building extraction from high spatial resolution remote sensing images via multiscale-aware and segmentation-prior conditional random fields," *Remote Sens.*, vol. 12, no. 23, p. 3983, Dec. 2020.

[31] Y. Chen, Y. Li, and J. Wang, "An end-to-end oil-spill monitoring method for multisensory satellite images based on deep semantic segmentation," *Sensors*, vol. 20, no. 3, p. 725, Jan. 2020.

[32] M. Krestenitis, G. Orfanidis, K. Ioannidis, K. Avgerinakis, S. Vrochidis, and I. Kompatsiaris, "Oil spill identification from satellite images using deep neural networks," *Remote Sens.*, vol. 11, no. 15, p. 1762, Jul. 2019.

[33] J. Zhang, H. Feng, Q. Luo, Y. Li, J. Wei, and J. Li, "Oil spill detection in quad-polarimetric SAR images using an advanced convolutional neural network based on SuperPixel model," *Remote Sens.*, vol. 12, no. 6, p. 944, Mar. 2020.

[34] Z. Jiao, G. Jia, and Y. Cai, "A new approach to oil spill detection that combines deep learning with unmanned aerial vehicles," *Comput. Ind. Eng.*, vol. 135, pp. 1300–1311, Sep. 2019.

[35] S. T. Yekeen, A. Balogun, and K. B. W. Yusof, "A novel deep learning instance segmentation model for automated marine oil spill detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 167, pp. 190–200, Sep. 2020.

[36] X. Yu, H. Zhang, C. Luo, H. Qi, and P. Ren, "Oil spill segmentation via adversarial f-divergence learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 4973–4988, Sep. 2018.

[37] L. Ding, H. Tang, and L. Bruzzone, "LANet: Local attention embedding to improve the semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 426–435, Jan. 2021.

[38] A. G. Roy, N. Navab, and C. Wachinger, "Concurrent spatial and channel 'squeeze & excitation' in fully convolutional networks," 2018, *arXiv:1803.02579*. [Online]. Available: https://arxiv.org/abs/1803.02579

[39] K. Song *et al.*, "Deep learning prediction of incoming rainfalls: An operational service for the city of Beijing China," in *Proc. Int. Conf. Data Mining Workshops (ICDMW)*, Nov. 2019, pp. 180–185.

[40] O. Oktay *et al.*, "Attention U-Net: Learning where to look for the pancreas," 2018, *arXiv:1804.03999*. [Online]. Available: https://arxiv.org/abs/1804.03999

[41] P. Zhao, J. Zhang, W. Fang, and S. Deng, "SCAU-Net: Spatial-channel attention U-Net for gland segmentation," *Frontiers Bioeng. Biotechnol.*, vol. 8, p. 670, Jul. 2020, doi: 10.3389/fbioe.2020.00670.

[42] *ALOS-PALSAR-Product-Format-Description-Level-1-1.5.pdf*. Accessed: Aug. 2, 2021. [Online]. Available: https://earth.esa.int/eogateway/documents/20142/37627/ALOS-PALSAR-Product-Format-Description-Level-1-1.5.pdf

[43] S. K. Chaturvedi, S. Banerjee, and S. Lele, "An assessment of oil spill detection using sentinel 1 SAR-C images," *J. Ocean Eng. Sci.*, vol. 5, no. 2, pp. 116–135, Jun. 2020.

[44] (Apr. 26, 2010). *Gulf Oil Spill Covers 817 Square Miles*. SkyTruth. Accessed: Mar. 22, 2021. [Online]. Available: https://skytruth.org/2010/04/gulf-oil-spill-covers-817-square-miles/

[45] (Aug. 15, 2017). *Satellite Imagery Reveals Scope of Last Week's Oil Spill in Kuwait*. SkyTruth, Accessed: Mar. 19, 2021. [Online]. Available: https://skytruth.org/2017/08/satellite-imagery-reveals-scope-of-last-weeks-massive-oil-spill-in-kuwait/

[46] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Las Vegas, NV, USA, Jun. 2016, pp. 770–778.

[47] W. Zhou, A. C. Bovik, H. R. Sheikh, and E. P. Simoncelli, "Image quality assessment: From error visibility to structural similarity," *IEEE Trans. Image Process.*, vol. 13, no. 4, pp. 600–612, Apr. 2004.

[48] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*. [Online]. Available: https://arxiv.org/abs/1412.6980

[49] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Med. Image Comput. Comput. Assist. Intervent (MICCAI)*, 2015, pp. 234–241.

[50] L. Zhou, C. Zhang, and M. Wu, "D-LinkNet: LinkNet with pretrained encoder and dilated convolution for high resolution satellite imagery road extraction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2018, pp. 182–186.

[51] L.-C. Chen, G. Papandreou, F. Schroff, and H. Adam, "Rethinking atrous convolution for semantic image segmentation," 2017, *arXiv:1706.05587*. [Online]. Available: https://arxiv.org/abs/1706.05587

[52] A. Chaurasia and E. Culurciello, "LinkNet: Exploiting encoder representations for efficient semantic segmentation," in *Proc. IEEE Vis. Commun. Image Process. (VCIP)*, Dec. 2017, pp. 1–4.

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